# JOSEFFUS SANTOS

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## EDUCATION

**University of Manitoba** — Winnipeg, MB Canada
Bachelor of Computer Science Honours Co-op DGPA 4.0 / 4.5 4th year — Expected May 2024
Bachelor of Science Major Biochemistry — Graduated

## TECHNICAL SKILLS

- TypeScript
- Node.js
- React
- Angular
- Ubuntu
- C#/.NET
- Fusion360
- Python
- OpenCV
- AWS/GCP
- Scikit
- Tensorflow 2
- Docker
- C / C++
- Embedded
- SQL
- REST
- Catch2
- PyTest
- Git

## EXPERIENCE

**Programmer Analyst** — Sep 2023 – Present
**Research Technician** — Feb 2011 – Sep 2023
Canadian Grain Commission, Government of Canada

- Led a team of 6 students in collaboration with the University of Manitoba to explore machine learning models to predict the incidence and severity of Ergot. Created ETL pipelines in Python to scrape weather data from ground station and satellite sources. Collected soil type data from open data sources for regions of interest. Stored the information in dockerized PostGres DB containers. Created dockerfiles to build custom development environments based on Nvidia Tensorflow docker images. Developed CI pipelines using GitHub Actions for automated type checking and linting. Followed Agile methods such as sprint planning, peer review, and weekly standups. Held biweekly progress meetings with stakeholders in the Government and the University of Manitoba.
- Developed programs in C# to extract data and add custom functions to instrument control software to speed up data analysis.
- Developed VBA scripts used across departments to automate data handling in Excel for the evaluation of research data.
- As part of an exploratory trial, AWS cloud administration. Designed and maintained a network of Windows and Linux EC2 instances to trial data analysis on the cloud with deployment scripts for users to automate service orchestration using AWS cli.
- Designed and fabricated an Atmel SAMD based robotic liquid sampling and dispensing system with firmware written in C++
- Gathered software requirements from stakeholders internal and external to the organization. Implemented new features based on user feedback.
- Contributed to data collection, analysis, and authoring research papers. Conducted journal searches and wrote review summaries of the current state of research on topics of interest.
- For four years, led task planning and scheduling for 4 technicians to organize work for the Harvest survey program.

**Software Developer** (Full Stack) Part-time Consultant — Sep 2023 – Dec 2023
**Software Developer** (Full Stack) Co-op — Jan 2023 – Apr 2023
AGCO Corporation

- Implemented new functional React components to display remote sensor data from LoRaWAN endpoints on a dashboard. Converted vanilla JavaScript class based React components to TypeScript functional React components.
- Implemented an alert system to warn users of critical fan system control failures to catch them early and prevent losses.
- Contributed new backend features in Python to support new API functionality using AWS Chalice for serverless AWS applications.
- Troubleshot and fixed bugs in both front end and backend code.
- Managed local dev environments in Docker containers for local dev and testing. Addressed Issues that came up on GitLab CI/CD pipeline tests to fix non conforming code. Created PyTest and Jest, unit and integration tests.
- Worked in an Agile environment, participated in scrum activities.

**Software Engineer** (Full Stack) Part-time Consultant — Mar 2023 – Present
DecisionWorks

- Designed and implemented a dashboard system to display and manage railway maintenance jobs with a frontend in Angular TypeScript, backend in C# dotnet core on an IIS server, and a data layer in MS SQL. Developed an embedded Google maps with GeoJSON layers, custom markers, and clusterers for data visualization and a calendar scheduling system to plan and deploy tasks.
- Created scripts to deploy AWS EC2 and lightsail resources using AWS cli, AWS EventBridge, AWS Lambda. Developed E2E automated integration tests using Playwright.
- Communicated with the client to gather user requirements and proposed solutions to address their needs. Estimated time and effort required to develop requested features. Incorporated feedback into future design and deliverables.

## PROJECTS

**Personal Cloud / Home Lab** | ZFS, Linux, GCP, AWS, Virtualbox, Docker 2015 - Present

- Network of multiple LAN hosted, cloud hosted, and hypervisor hosted web services
- Mixture of LAN Linux and Windows hosts; Linux hosts on AWS that run applications in docker to provide VPN, Reverse Proxy, SOCKS5, SSH, remote desktop, content delivery, server monitoring and other services.

**Forest Fire Prediction Model** | Python, Docker, PostGres, QGIS, GeoPandas, sklearn, Tensorflow 2022

- Used Python to scrape data from weather stations nearest forest fires
- Extended PostGres and Nvidia Tensorflow containers to build dev environments for machine learning
- Trained and evaluated SVM and Random Forest models for forest fire severity prediction using weather

**Meal Planner Calorie Tracking App** | Android, Java, JUnit, Espresso, HSQLDB 2022

- Designed and built a Java Android app following Agile development practices
- Applied software design patterns, implemented object oriented class design, and thorough unit testing
- Implemented a relational database in Hyper SQL DB to store application data. Followed a 3 layer architecture

**Piston Pump** | Fusion360, KiCAD, C++, Atmel SAMD 2019 - 2022

- Designed and built a piston based liquid delivery device that exceeded ASTM class A standards for accuracy and delivery speed
- Designed touchscreen interface for human interaction, control, calibration, error recovery

**City Winnipeg Database** | Python, Qt, SQLite, Google Maps API, GeoPandas, Pandas 2021

- Combined multiple open data sources about the City of Winnipeg into a 3NF and BCNF normalized relational database
- Used Google Maps API to get asset geographic coordinates and used GeoPandas to measure distances and regional inclusions
- Created Windows, Linux cross platform GUI program in Python to run queries against SQLite DB

**Fuzzy Video Content Matcher** | C/C++, Python, Ubuntu 2020 - 2022

Perceptual hashing to find similar videos. Implemented a python script on Linux to use Facebook TMK+PDQF algorithm

**3D Printer** | Fusion360, C++, RAMPS, Arduino Mega, RaspberryPi 2016 - 2017

Designed and built an FDM 3D printer with control and monitoring via web interface

**Wireless Controllable Outlet Hub** | C/C++, HTML, CSS, JS, esp8266 2016

Lan/Wifi connected hub for controlling remote switches over 433mhz with cross platform embedded web page for control

## EXTRACURRICULAR

**University Drone Club** | Git, C/C++, Python, OpenCV, Qgroundcontrol 2022 - 2023

- Created QR code reader to parse drone waypoints, and GPS boundary points to avoid
- Developed algorithm to avoid a Geo boundary and find a path to the next waypoint
- Developed system to build json files with updated Qgroundcontrol mission and upload the new mission via MAVLink
- Created AI subsystem to autonomously identify people, measure distance, and track movement from an airborne drone

**University Dev Club Hackathon |** Git, Python, OpenCV, Mediapipe 2023

Implemented gesture recognition on top of the MediaPipe ML library to create a hand tracking mouse in 12hrs

**University Biomedical Engineering club Gaming Head Mouse** | C/C++, Mbed OS 2021 - 2022

Created a head mounted gyroscopic joystick to allow a disabled gamer play with head movements

**Magpie CTF Hackathon**| Ghidra, IDA Free, Ollydbg 2021

Participated in a capture the flag hackathon involving reverse engineering and penetration testing

## HOBBIES

**Obstacle Course Racing**

Trained and participated in 6 5K, 10K, 21K Spartan races , Attended Tough Mudder Twice , a few Mud Hero's

**Rock Climbing**

**Lunch Ball Hockey and Tennis**